

National University of Computer and Emerging Sciences
Lahore Campus

Computer Networks (CS3001)

Date: May 19th, 2026

Course Instructor(s)

Ms. Sarah Khaleel, Mr. Nauman Moazzam Hayat

Final Exam

Total Time: 3 Hours

Total Marks: 90

Total Questions: 8

Roll No

Section

Student Signature

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- Instruction/Notes:**
- Attempt all questions on the provided separate answer sheet, except **Question 2, 4, 5 & 8** which are to be answered on the **Question Paper**, to be attached with the answer sheet.
 - Clearly write corresponding question number and part number at the top center of the answer sheet with a thick pen / marker before starting a new question / answer.
 - In case you have used rough sheets, they should **NOT** be attached to the answer sheet.
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CLO 1 (Question 1): Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Q1: Answer the following MCQs by **encircling** the correct option on the question paper: Any erasing, cutting and overwriting is not allowed. (Answers given in the answer sheet will **not be marked**.) [1 * 6 = 6 Marks]

1. A packet of $L = 12,000$ bits is sent across 3 links in series (store-and-forward). Each link has $R = 2$ Mbps & propagation delay = 8 ms. Processing and queuing delays are negligible. What is the total end-to-end delay?

- A. 18 ms B. 42 ms C. 24 ms D. 34 ms

Answer: B

2. Which type of DNS query puts more load at the upper levels of the DNS Server hierarchy?

- A. Recursive B. Iterative C. Root D. TLD

Answer: A

3. Why even bother to have UDP while TCP exists?

- A. UDP shouldn't exist B. Legacy C. UDP is reliable D. UDP is faster

Answer: D

4. A 2000 bytes IPv6 datagram will be fragmented into _____ fragments by a router with an MTU of 500 bytes:

- A. 4 B. 5 C. 6 D. 0

Answer: D

5. BGP is classified as which protocol:

- A. Link State B. Distance Vector C. Path Vector D. Hybrid

Answer: C

6. The three categories of MAC protocols are:

- A. Channel Partitioning, Random Access, Taking Turns B. Intra AS, Inter AS, Non AS
C. Fast, Slow, Mixed D. Pipelined, Stop & Wait, Hybrid

Answer: A

CLO 2 (Questions # 2, 3, & 4): Demonstrate the basics of network concepts using state-of-the-art network tools/techniques.

Q2: A message of $M = 10$ Megabits must be sent from Host A to Host C through an intermediate router B. Link A→B: $R_1 = 2$ Mbps, $d_{prop1} = 20$ ms. Link B→C: $R_2 = 1$ Mbps, $d_{prop2} = 30$ ms. Store-and-forward packet switching is used with packet size $L = 2000$ bits (including all headers). Ignore processing & queuing delays. [2 + 5 = 7 Marks]

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- (a) How many packets does the message produce?
(b) Calculate the total end-to-end delay i.e. from Host A to Host C for the complete message WITH pipelining carefully.

Sol:

(a) Number of packets

$$N = M/L = 10 \times 10^6 / 2000 = 5000 \text{ packets}$$

(b) Total delay (with pipelining)

Transmission delays:

$$tx1 = 2000 / 2 \times 10^6 = 1 \text{ ms}, \quad tx2 = 2000 / 1 \times 10^6 = 2 \text{ ms}$$

Propagation:

$$d_1 = 20 \text{ ms}, \quad d_2 = 30 \text{ ms}$$

First packet delay:

$$(1 + 20) + (2 + 30) = 53 \text{ ms}$$

Bottleneck link: $tx2 = 2 \text{ ms}$

Remaining packets:

$$(5000 - 1) \times 2 = 9998 \text{ ms}$$

Total delay:

$$53 + 9998 = 10051 \text{ ms} \approx 10.051 \text{ s}$$

Q3: A web browser fetches a web page that consists of 1 base HTML file and 8 embedded image objects. The link transmission rate is $R = 10 \text{ Mbps}$. The base HTML is 20 KB and each image is 50 KB. $RTT = 40 \text{ ms}$. Ignore queuing and processing delays. **[2 + 3 + 2 = 7 Marks]**

- (a) Calculate total download time using non-persistent HTTP/1.0 (no parallel connections).
(b) Calculate total download time using non-persistent HTTP/1.0 with 5 parallel connections.
(c) Calculate total download time using persistent HTTP/1.1 with pipelining.

Ans:

(a) Non-persistent HTTP/1.0 (no parallel)

$$\text{Transmission for Base} = 20 \times 10^3 \times 8 / 10 \times 10^6 = 16 \text{ ms}$$

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Transmission for each image = $50 \times 10^3 \times 8 / 10 \times 10^6 = 40 \text{ ms}$

Each object = 2 RTT + transmission

- Base HTML: $2(40) + 16 = 96 \text{ ms}$
- 8 images: $8 \times [2(40) + 40] = 960 \text{ ms}$

Total = $96 + 960 = 1056 \text{ ms} \approx 1.056 \text{ s}$

(b) Non-persistent HTTP/1.0 (5 parallel connections)

- Base HTML: 96 ms

Images in batches of 5:

- Batch 1 (5 images):
 $2\text{RTT} + \text{transmission} = 80 + 40 = 120 \text{ ms}$
- Batch 2 (remaining 3 images):
120 ms

Total = $96 + 120 + 120 = 336 \text{ ms} \approx 0.336 \text{ s}$

Part (c) Persistent HTTP/1.1 with pipelining:

With persistent HTTP/1.1 + pipelining, one TCP connection is opened and reused. The client sends all requests back-to-back without waiting for responses.

TCP handshake (1 RTT = 40 ms) + GET base HTML (1 RTT + transmission = $40 + 16 = 56 \text{ ms}$) + Browser pipelines all 8 image GETs simultaneously (Total transmission for all 8 images = $8 \times 40 \text{ ms}$ (320) + 1 RTT for the round trip = $40 + 320 \text{ ms} = 360 \text{ ms}$)

Total = $40 + 56 + 360 = 456 \text{ ms}$

Q4: Host A and Host B are communicating using TCP. The Maximum Segment Size (MSS) is 1 Kbyte (1000 bytes.) Host A's initial sequence number is 5000. Host A has a large amount of data to send and is currently in the Congestion Avoidance phase with a Congestion Window (cwnd) of 20 Kbytes & a Threshold (ssthresh) of 32 Kbytes. Host A transmits a full window of data (i.e. 20 segments) in a single burst, starting with Segment 1 having starting Sequence Number of 5000. Answer the following: **[3+1+2+2+1+1+5 = 15 Marks]**

- What are the Sequence Numbers for Segments 1, 2 & 3 in this burst? **5000, 6000, 7000**
- Assume Segment 2 is lost but all other 19 segments (Segment 1 and Segments 3 till 20) arrive successfully and in sequence at Host B. What is the Acknowledgment Number contained in the packet sent by Host B in response to the arrival of Segment 3? **6000**
- If the loss of segment 2 had been detected via Triple Duplicate ACKs, what will be the new values for ssthresh and cwnd (in bytes) if it was TCP Tahoe? **ssthresh = cwnd / 2 = 20 KB / 2 = 10 KB, cwnd = 1 KB**

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- (d) If the loss of segment 2 had been detected via Triple Duplicate ACKs, what will be the new values for ssthresh and cwnd (**in bytes**) if it was the newer variant of TCP Reno? $ssthresh = cwnd / 2 = 20 \text{ KB} / 2 = 10 \text{ KB}$, $cwnd = ssthresh + 3 \text{ MSS} = 10 \text{ KB} + 3 \text{ KB} = 13 \text{ KB}$
- (e) If the loss of segment 2 had been detected via Timeout, what will be the new cwnd (in bytes) for TCP Tahoe? **1 KB**
- (f) If the loss of segment 2 had been detected via Timeout, what will be the new cwnd (in bytes) for TCP Reno? **1 KB**
- (g) Assuming TCP Reno again & loss of segment 2 was detected via three duplicate ACKs, Host A enters the Fast Recovery state. As the remaining segments from the original 20-segment burst (Segments 4 till segment 20) continue to arrive in order at Host B, they keep on triggering further duplicate ACKs (in addition to the four same ACKs already received (i.e. one original ACK plus three duplicate ACKs.) How many further duplicate ACKs must Host A receive for its cwnd to reach 18 KB? **5 further duplicate ACKs.** (cwnd at start of Fast Recovery is 13 KB, each further duplicate ACK increases the cwnd by 1 MSS every RTT, thus when cwnd is 18 KB, then $18 - 13 = 5 \text{ KB}$, meaning 5 further duplicate ACKs.)

Q5 (a): An organization having a CIDR IP address block of 200.10.128.0/20 has three subnets connected via a single router. The three subnets must be contiguously assigned starting from the beginning of the parent block as per the below requirements: **[2 * 6 = 12 Marks]**

- Subnet A requires 1000 host addresses
- Subnet B requires 500 host addresses
- Subnet C requires 100 host addresses

Answer the following in dotted decimal notation including prefix (answers in binary will not be marked):

- i) Subnet address of Subnet A: **200.10.128.0/22**
- ii) Broadcast address of Subnet A: **200.10.131.255/22**
- iii) Subnet address of Subnet B: **200.10.132.0/23**
- iv) Broadcast address of Subnet B: **200.10.133.255/23**
- v) Subnet address of Subnet C: **200.10.134.0/25**
- vi) Broadcast address of Subnet C: **200.10.134.127/25**

Q5 (b): Refer to the router in part (a) above, it has four interfaces as given below:

- Interface 0: Connected to Subnet A
- Interface 1: Connected to Subnet B
- Interface 2: Connected to Subnet C
- Interface 3: Connected to the internet (Default)

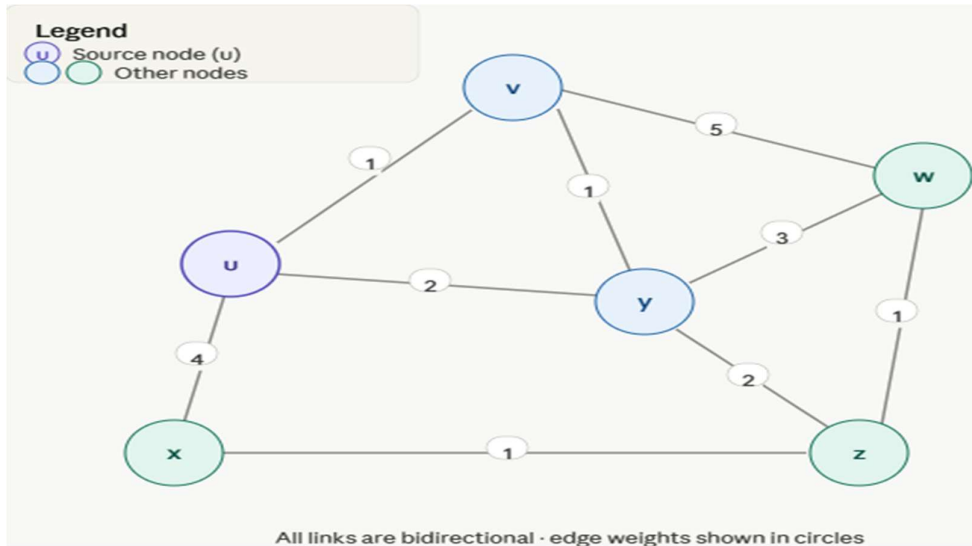
Construct the forwarding table of this router as per the below format. The default entry is given below for reference. **[3 Marks]**

Destination Prefix (Dotted Decimal Notation)	Interface ID
* (Default)	3
200.10.128.0/22	0
200.10.132.0/23	1
200.10.134.0/25	2

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Q6: Consider the network graph below (all links bidirectional):

[14 + 6 = 20 Marks]



(a) Run Dijkstra's algorithm for node **u** & fill in the below table showing each step.

Step	N'	v D(v) , P(v)	w D(w) , P(w)	x D(x) , P(x)	y D(y) , P(y)	z D(z) , P(z)
0	u	<u>1, u</u>	∞	4, u	2, u	∞
1	uv		6, v	4, u	<u>2, u</u> or <u>2, v</u>	∞
2	uvy		5, y	<u>4, u</u> or 4, u		4, y or <u>4, y</u>
3	uvyx or uvyz		5, y or 5, z	----- or 4, u		4, y or -----
4	uvyxz or uvyzx		<u>5, y</u> or <u>5, z</u>			
5	uvyxyzw or uvyxyzw					

(b) Provide the forwarding table for router **u**. (1st entry has been prefilled for reference.)

Destination	Outgoing Link	Cost
v	(u , v)	1
w	(u , y) or (u , v)	5
x	(u , x)	4
y	(u , y) or (u , v)	2
z	(u , y) or (u , v)	4

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Q7(a): For Distance vector routing in three nodes have a linear topology, i.e. A ----- B ----- C, with (A ----- B) link cost = 1 & (B ----- C) link cost = 1. Node C goes down. B is going to update $D_B(C) = \infty$ but before doing this, B receives $D_A(C) = 2$ from A. What does B compute as the value for $D_B(C)$ in this scenario? **Three (3)** [3 Marks]

Q7(b): For part (a) what is this issue called? What is the name of the solution to this issue? [2 Marks]

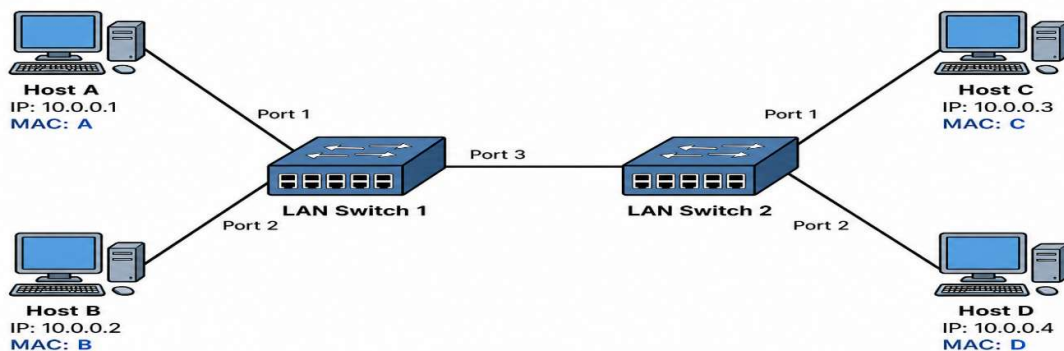
Solution: Count to Infinity Problem. Solution is called Poison Reverse or Split Horizon.

Q7(c): Cyclic Redundancy Check (CRC) can be used for error detection at the link layer. The sender and receiver have agreed on a generator polynomial G, represented by the bit string **10101**. Answer the following questions: [1 + 2 + 2 = 5 Marks]

- How many redundant bits (r) will be appended to the data payload before transmission? **5 - 1 = 4 bits**
- Let D be the data payload and R be the CRC bits. The sender transmits the combined bitstring i.e. $\langle D, R \rangle$. If the transmission is **error-free**, what should be the remainder left at the receiver after performing modulo-2 division? **Zero (0)**
- Refer to the hardware at the link layer, why is CRC generally preferred over the "Internet Checksum" even though CRC is mathematically more complex to calculate? (Brief answer.)
Solution: Better / stronger error detection by CRC than Checksum, as checksum may miss errors, specially burst errors. (Optional additional hardware related reason can be that CRC is more efficient than checksum for hardware implementation.)

Q8: Refer to the diagram below, all host ARP caches and switch forwarding tables are initially empty. Consider the following *chronological* sequence of events:

- At Time **T1**: Host A sends an ICMP Echo Request (Ping) to Host B.
- At Time **T2**: Host C sends an ICMP Echo Request (Ping) to Host A.
- At Time **T3**: Host B sends an ICMP Echo Request (Ping) to Host C.



Answer the following:

[2 + 2 + 6 = 10 Marks]

- At T1, does an ARP Request originate? If yes, which node(s) generates the request and which node(s) replies to the request? **Yes. A generates an ARP request for 10.0.0.2 & B replies.**
- At T2, does an ARP Request originate? (Yes/No only). **Yes.**

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- c) Immediately after Event T3 completes (Request and Reply), fill in the **Switch Forwarding Tables** for **S1** & **S2** given below.

Switch Table: S1

MAC Address	Port
A	1
B	2
C	3

Switch Table: S2

MAC Address	Port
C	1
A	3
B	3